

Contact

cordova.nellie@outlook.com

www.linkedin.com/in/cordovank
(LinkedIn)

cordovank.github.io/ (Portfolio)

Top Skills

Gradio

Multi-agent Systems

Deep Learning

Languages

English (Professional Working)

Catalan (Elementary)

Spanish (Native or Bilingual)

Certifications

Microcomputer Systems and Network

Docker in Depth

AWS Certified AI Practitioner

Group 2 Social / Behavioral Research Investigators and Key Personnel

Applied Machine Learning: Algorithms

Honors-Awards

Phi Theta Kappa

Pi Mu Epsilon

Nellie K. Cordova

ML Engineer | RAG Systems · Agentic AI · NLP | Georgia Tech
OMSCS | Python · PyTorch · FastAPI
United States

Summary

I build ML and AI systems that are designed to be evaluated, not just demonstrated. My recent work focuses on retrieval-augmented generation (RAG), agentic workflows, and multimodal pipelines — combining production engineering discipline with graduate-level ML training from Georgia Tech's OMSCS program (ML specialization).

On the applied side: I've built a modular RAG framework with hybrid retrieval (BM25 + FAISS, RRF fusion, MMR reranking), explicit wiring, and observability-first architecture. I've fine-tuned Vision Transformers and GPT-2 for a multimodal food-to-recipe system. I've designed schema-first agentic pipelines with CrewAI where the reliability contract — no hallucination, QA-audited outputs — is a first-class constraint, not an afterthought.

Earlier in my career, I spent two years at JPMorgan building Java microservices on a real-time payments platform: strict API contracts, JUnit/Cucumber test automation, CI quality gates, and regulated release readiness. That engineering background is how I approach ML systems — evaluable by design, reliable enough to ship. I'm looking for ML Engineer or AI Engineer roles where those two things matter together.

Portfolio: cordovank.github.io

Experience

JPMorgan Chase & Co.

2 years 4 months

Software Engineer (Payment Service)

June 2019 - September 2021 (2 years 4 months)

- Contributed end-to-end across the SDLC (requirements to deployments) for Java-based payments microservices and drove production readiness.

- Authored comprehensive operational runbooks—architecture, dependencies/configs, failure modes, diagnostic checklists, decision trees, and step-by-step remediation—to enable fast triage, recovery, and self-serve on-call; provided light support on monitoring/alerting dashboards with platform/SRE.
- Built and maintained test automation (JUnit, Cucumber); enforced CI quality gates (tests, static analysis, coverage) to keep merges green, and executed functional/perf testing (Postman, SOAP UI, JMeter).
- Managed regulated releases: compiled release evidence (test/coverage reports, change tickets, approvals) to secure production sign-off; presented services to global production management.
- Led/participated in agile rituals (sprint planning, backlog refinement); authored technical/architecture docs; coordinated across time zones.

Developer (Salesforce)

September 2019 - June 2020 (10 months)

Participated in building a sustainable software solution in an agile discipline to deliver a socially focused organization's strategy more efficiently. Successfully enabled the client to collect data in a single consistent fashion using Salesforce and facilitating reporting and metrics.

- Gathered requirements from stakeholders, translated them into technical changes, implemented the updates, and documented the work.
- Enhanced a custom Visualforce UI for clarity and consistency, and automated survey creation with declarative Flows to reduce manual setup and standardize templates.

William Paterson University of New Jersey

1 year 8 months

Math & CS Tutor

September 2018 - May 2019 (9 months)

Academic Success Center

- Tutored undergraduate students in math (statistics, college algebra, calculus, and other higher level math) and computer science (Object-Oriented Programming: C++).

ML Research Assistant, Data Science Research Project

March 2018 - May 2019 (1 year 3 months)

Computer Science Department

- Project: A Knowledge Discovery Approach to Analyzing Mental Health Problems in NYC

- Implemented and evaluated predictive models (KNN, Logistic Regression, clustering) in Python & R to make inferences about mental health behaviors among teenagers who tan.
- Presented research insights to students and faculty members during WPU Explorations conference week.

Student Office Assistant

October 2017 - May 2018 (8 months)

College of Education

- Provided excellent customer service to diverse group of clients, including students, employers, and visitors by answering phone calls, emails, and in-person inquiries.
- Compiled student comments from formal end-of-semester evaluations into one document and convert it into an electronic document for each faculty and staff in the department.
- Maintained a hard copy backup of all sensitive information keeping accurate records.
- Assisted with other projects as assigned.

Union County College

Student Office Assistant @ Career Services Center

March 2015 - July 2016 (1 year 5 months)

Elizabeth, NJ

- Developed promotional material for the Career Services Center and assisted with other projects as assigned.
- Trained and mentored new student employees and clarified performance expectations - organized work schedules, set goals, and monitored progress.
- Provided meaningful data entry for job postings.
- Assisted administrators with the marketing of annual job fair.
- Assisted students with career inquiries and/or financial aid application process.
- Provided front-desk reception and other clerical office support as required.
- Updated job knowledge by participating in educational opportunities.
- Maintained excellent positive customer/student relations.

Education

Georgia Institute of Technology

Master's degree, MS Computer Science - Machine Learning · (2022 - 2025)

William Paterson University of New Jersey

Bachelor of Arts (B.A.), Mathematics and Computer Science · (2016 - 2019)

Union County College

Associate of Arts (A.A.), Education (Grades 4-12) · (2014 - 2016)

IES San Juan - Seville, Spain

Technician in Microcomputer Systems and Networks, Computer Systems

Networking and Telecommunications · (2009 - 2011)